



Department of Computer Science and Engineering (Data Science)

Subject: Machine Learning – I (DJS23DSL402)

AY: 2025-26

Experiment 8

(Association Rule Mining)

Aim: Explore Association Rule Mining with variations on different datasets.

Theory:

Association Rule Mining:

Association Rule Mining is a machine learning method meant to discover interesting relationships, frequent patterns, associations, or correlations among a set of items in transactional or relational data. This technique is commonly used in market basket analysis where one seeks to find associations between products bought together.

It is composed of two important steps:

- **Frequent Itemset Generation:** Identify all itemsets whose support is greater than or equal to the minimum support threshold.
- **Rule Generation:** Extract rules that meet minimum confidence from the frequent itemsets.

Important Terminologies:

- **Support:** The proportion of transactions that contain the itemset.
- **Confidence:** Indicates the number of times the if-then statements are found true.
- **Lift:** Ratio of observed support to expected support if X and Y were independent.

Popular algorithms include Apriori and FP-Growth:

- Apriori uses a breadth-first search strategy and generates candidate itemsets of length k from itemsets of length k-1.
- FP-Growth avoids the costly candidate generation and uses a prefix-tree (FP-tree) structure for storing compressed data.

Lab Assignments to complete in this session:

Use this Dataset:

```
[['Milk','Onion','Nutmeg','Kidney Beans','Eggs','Yogurt'],  
 ['Dill','Onion','Nutmeg','Kidney Beans','Eggs','Yogurt'],  
 ['Milk','Apple','Kidney Beans','Eggs'],  
 ['Milk','Unicorn','Corn','Kidney Beans','Yogurt'],  
 ['Corn','Onion','Onion','Kidney Beans','Ice cream','Eggs']]
```



Task 1:

- Generate frequent itemsets using the Apriori algorithm with a minimum support of 0.6.
- Display the frequent itemsets along with their support values.

Task 2:

- Modify the minimum support threshold to 0.7.
- Regenerate the frequent itemsets.
- Compare the results with Task 1.

Task 3:

- Generate association rules from the frequent itemsets
- Display the association rules with their confidence and lift values.

Task 4:

- Explore the impact of different confidence thresholds on the generated association rules.